

CNN-Based Purple Blotch Disease Detection in Onion Crops Using Lightweight Deep Learning Architectures

Project ID: R-26-IT-100

Project Proposal Report

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January 2026

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at

any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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The above candidate is conducting research for their undergraduate Dissertation under my supervision.

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ABSTRACT

Onion (*Allium cepa*) cultivation is a vital agricultural activity in Sri Lanka, contributing significantly to national food security and rural livelihoods. However, onion crops are widely susceptible to purple blotch disease caused by *Alternaria porri*, a devastating foliar fungal pathogen that can reduce yields by 30–50% under favorable environmental conditions. Current disease identification methods rely predominantly on subjective visual field inspections and delayed expert consultations, resulting in untimely or incorrect diagnoses, excessive fungicide usage, and economic losses for smallholder farmers.

This research proposes the design, development, and validation of a CNN-based mobile disease detection system for real-time identification and severity classification of purple blotch in onion crops under Sri Lankan field conditions. The system leverages lightweight deep learning architectures—EfficientNet, MobileNetV3, and Vision Transformers—fine-tuned on a locally collected dataset of 3,000+ annotated onion leaf images from FCRDI demonstration plots. AutoML via FLAML automates architecture search and hyperparameter optimization, while TinyML techniques (INT8 quantization and pruning) enable deployment as a sub-10 MB TensorFlow Lite model for offline inference on resource-constrained mobile devices.

The expected outcome is a high-accuracy (>90%), mobile-optimized disease detection model capable of classifying onion leaves into four categories (Healthy / Early / Mid / Severe), providing severity scores (0–100%), and recommending appropriate fungicides and cultural control practices. This system directly addresses the critical gap in locally validated, crop-specific AI models for Sri Lankan agriculture, empowering farmers with autonomous, real-time diagnostic capability at the field level.

Keywords: Purple Blotch, *Alternaria porri*, CNN, EfficientNet, MobileNet, Vision Transformer, TinyML, AutoML, Onion Disease Detection

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my dissertation supervisor Dr. Kaushalya Dissanayake, and co-supervisor Dr. Dharshana Kasthurirathna, for their valuable guidance and support throughout the research process. Their perceptive comments and helpful critique have been instrumental in refining my research and improving the overall quality of my work.

I would also like to extend special thanks to Dr. W.M.K Fernando, Principal Agriculture Scientist (Plant Pathology) at the Field Crops Research and Development Institute (FCRDI), for providing expert domain guidance on onion crop pathology, purple blotch biology, and disease severity assessment protocols.

I am grateful to the staff of the Faculty of Computing for their assistance and encouragement throughout the course of my studies. Their dedication to teaching and research has been an inspiration to me.

I am also thankful to my fellow team members for their support and collaboration. Our discussions and exchanges of ideas have enriched my understanding of the subject matter. Their contributions have been instrumental in making this research possible.

Finally, I would like to express my appreciation to the onion farming

communities and FCRDI field officers who have participated in this study and contributed their time and insights. Their willingness to share their experiences and allow image collection on their farms has been invaluable in building a representative, locally relevant disease dataset.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
CNN	Convolutional Neural Network
ViT	Vision Transformer
TFLite	TensorFlow Lite
AutoML	Automated Machine Learning

FLAML	Fast and Lightweight AutoML
TinyML	Tiny Machine Learning
FCRDI	Field Crops Research and Development Institute
DOA	Department of Agriculture
FAO	Food and Agriculture Organization
SDG	Sustainable Development Goal
IPM	Integrated Pest Management
GDD	Growing Degree Days
INT8	8-bit Integer Quantization
mAP	Mean Average Precision
RGB	Red Green Blue
HSV	Hue Saturation Value
Grad-CAM	Gradient-weighted Class Activation Mapping
API	Application Programming Interface
UI	User Interface
WBS	Work Breakdown Structure

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1. INTRODUCTION

1.1 Background and Context

Onion (*Allium cepa*) cultivation is one of the most economically significant agricultural activities in Sri Lanka, contributing substantially to both national food security and the livelihoods of rural farming communities. According to the Department of Agriculture Sri Lanka, onion is cultivated across multiple agro-ecological zones, with the Maha Illuppallama region being a primary production hub. Despite its importance, onion farmers face persistent and multifaceted challenges that severely impact crop health and economic returns.

The Real-World Problem:

Sri Lankan onion crops are widely affected by purple blotch disease caused by the fungal pathogen *Alternaria porri*, which produces characteristic purple-brown concentric lesions on onion leaves and flower stalks [1]. Under favorable conditions of high humidity and warm temperatures—conditions prevalent across Sri Lanka's major onion-growing Dry Zone—purple blotch can cause yield losses of 30–50% [2]. Concurrent challenges including pest infestations from Thrips (*Thysanoptera*) and onion caterpillars (*Spodoptera exigua*), and nutrient deficiencies in nitrogen, potassium, and phosphorus further compound crop vulnerability [3]. Current disease identification relies on subjective visual field inspections by farmers or delayed visits from government agricultural officers, leading to frequent misdiagnosis, untimely fungicide application, and economic losses.

Who Is Affected and Why It Matters:

Sri Lanka has an estimated 150,000+ smallholder onion farming households. These farmers typically operate on small land parcels with limited resources and have constrained access to plant pathology expertise. Misidentification of purple blotch—often confused with Stemphylium blight, downy mildew, or nutrient deficiency symptoms—leads to inappropriate chemical applications, increased production costs, environmental degradation, and reduced yields. The Food and

Agriculture Organization (FAO) reports that post-harvest losses in onion crops in developing countries can reach 30–40%, much of which is attributable to preventable diseases [4].

Statistical Evidence:

- Sri Lanka imports approximately 200,000 metric tons of onions annually, highlighting a significant domestic production shortfall [5].
- Studies indicate that timely and accurate disease management could increase onion yields by 25–35% in the South Asian context [6].
- Only 12% of Sri Lankan farming households have access to digitalized agricultural advisory services [3].
- Purple blotch is reported in nearly all onion-growing regions of Sri Lanka, with peak severity during Maha and Yala seasons [1].

Analysis of Existing Solutions and Their Limitations:

Current plant disease detection applications, including Plant.id, Leafsnap, and Plantix, employ CNN-based image recognition for multi-crop disease classification. However, these platforms suffer from critical limitations for the Sri Lankan onion farming context: (1) they are trained on generic or foreign datasets that do not represent local crop varieties, climate conditions, and cultivation practices, resulting in poor field accuracy; (2) they offer only English-language outputs without localized disease management recommendations; (3) they lack offline functionality essential for rural farming areas with limited internet connectivity; and (4) they do not provide severity-graded classification or crop-specific fungicide recommendations [7].

The Proposed Solution:

This research proposes a CNN-based mobile disease detection system for real-time purple blotch identification and severity classification in onion crops. The system utilizes lightweight deep learning architectures—EfficientNet, MobileNetV3, and Vision Transformers—fine-tuned on a locally collected, expert-annotated dataset of 3,000+ onion leaf images from FCRDI demonstration plots. AutoML via FLAML automates architecture search across candidate models, while TinyML techniques (INT8 quantization and pruning) produce a compact (<10 MB) TensorFlow Lite model for offline inference on smartphones. The overall

platform comprises four integrated components:

1. **Component 1:** CNN-Based Purple Blotch Disease Detection (Member 1 – IT22054890) (*This component*)
2. **Component 2:** Pest Detection & Identification using YOLOv8 (Member 2 – IT22226464)
3. **Component 3:** Multilingual Agro-Advisory Chatbot – Sinhala/English/Singlish (Member 3 – IT22087256)
4. **Component 4:** Image-Based Nutrient Deficiency Detection (Member 4 – IT22142528)

SDG Alignment:

This project directly contributes to:

- **SDG 2 (Zero Hunger):** By enabling early, accurate disease detection that protects crop yields and reduces food production losses.
- **SDG 12 (Responsible Consumption and Production):** By reducing excessive and inappropriate fungicide application through data-driven diagnosis.
- **SDG 13 (Climate Action):** By promoting sustainable, precision-agriculture practices that minimize chemical inputs and environmental impact.

Research Cluster Alignment:

This research is aligned with the Autonomous Intelligent Machines and Systems (AIMS) research cluster and the Agri-Technology industry vertical, focusing on intelligent perception, decision-making, and human-centric AI solutions.

1.2 Literature Review

A comprehensive literature review was conducted examining recent scholarly works in CNN-based plant disease detection, lightweight model architectures, transfer learning for agricultural applications, and AutoML/TinyML deployment strategies. The following key findings emerged:

1. CNNs for Plant Disease Classification:

Mohanty et al. (2016) demonstrated the effectiveness of deep CNNs for plant disease identification using the PlantVillage dataset, achieving over 99% accuracy with GoogLeNet and AlexNet architectures on 38 disease classes across 14 crop species [8]. However, their models were trained under controlled laboratory conditions with uniform backgrounds, and performance degraded significantly under real-world field conditions with variable lighting, complex backgrounds, and co-occurring symptoms.

2. EfficientNet and Mobile-Optimized Architectures:

Tan and Le (2019) introduced EfficientNet, a family of CNN models that achieve state-of-the-art accuracy with significantly fewer parameters through compound scaling of depth, width, and resolution [9]. EfficientNet-B0 achieves comparable accuracy to much larger models while requiring 5.3M parameters—making it ideal for mobile deployment. MobileNetV3 (Howard et al., 2019) further optimizes for mobile inference using neural architecture search, squeeze-and-excitation blocks, and h-swish activation [10].

3. Vision Transformers for Agricultural Applications:

Dosovitskiy et al. (2021) introduced Vision Transformers (ViT), demonstrating competitive image classification performance using self-attention mechanisms rather than convolutional operations [11]. Recent agricultural applications show ViTs outperforming CNNs on leaf disease datasets when sufficient training data and augmentation are available, with particular strength in capturing global context and long-range dependencies in disease symptom patterns.

4. Transfer Learning for Limited Agricultural Datasets:

Dutta et al. (2022) reviewed the application of pre-trained CNN models for onion disease detection, finding that transfer learning from ImageNet significantly improves performance when local disease datasets are small [12]. Fine-tuning the final layers of pre-trained models on domain-specific data achieved over 90% accuracy in distinguishing *Alternaria porri* symptoms under controlled conditions.

5. AutoML for Architecture Selection:

Wang et al. (2021) introduced FLAML (Fast and Lightweight AutoML), an efficient framework that automates model selection and hyperparameter tuning with low computational cost [13]. Applied to

image classification, AutoML can systematically compare CNN architectures and training configurations, identifying optimal models without exhaustive manual experimentation.

6. TinyML for Edge Deployment:

Banbury et al. (2021) surveyed TinyML techniques for deploying deep learning models on resource-constrained devices [14]. Post-training INT8 quantization typically reduces model size by 4x with minimal accuracy loss (<1%), while magnitude-based pruning removes redundant weights, enabling real-time inference on smartphones without cloud connectivity.

7. Existing Plant Disease Detection Systems and Their Limitations:

Commercial applications like Plant.id, Plantix, and Agrio offer AI-driven crop health diagnostics but are limited to generic multi-crop models trained on international datasets. They do not account for Sri Lankan onion varieties, local environmental conditions, or disease severity grading. Fernando (n.d.) documented the specific challenges of managing onion purple blotch in Sri Lanka, emphasizing the need for tools adapted to local conditions [7]. The DOA's current advisory relies on static pamphlets and periodic field visits, lacking real-time, scalable diagnostic support.

8. Explainability in Agricultural AI:

Selvaraju et al. (2017) introduced Grad-CAM, a technique for generating visual explanations by highlighting image regions most influential in CNN predictions [15]. In agricultural disease detection, Grad-CAM provides critical transparency by showing farmers and pathologists which leaf areas drove the diagnosis, building trust in AI recommendations and enabling human verification.

1.3 Research Gap

The comprehensive literature review reveals several critical gaps that this research aims to address:

1. Absence of Locally Validated Purple Blotch Detection Models:

Existing CNN models for plant disease detection are predominantly

trained on generic datasets such as PlantVillage, collected under controlled laboratory conditions in North America and Europe. No CNN model has been specifically trained and validated for detecting *Alternaria porri* (purple blotch) on Sri Lankan onion varieties under tropical field conditions, where disease presentation is influenced by local climate, soil, and cultivar characteristics [7].

2. Lack of Multi-Level Severity Classification:

Current disease detection systems typically provide binary classification (healthy vs. diseased). For effective disease management, farmers need graded severity assessment (Early / Mid / Severe) that enables timely, proportionate responses. No existing system provides four-level severity classification with corresponding severity scores (0–100%) specifically for onion purple blotch.

3. No AutoML-Optimized Architecture Selection for Onion Diseases:

While multiple CNN architectures have been applied to plant disease detection, the optimal choice for onion purple blotch under field conditions is unknown. No study has systematically compared EfficientNet, MobileNetV3, and Vision Transformers using AutoML-driven architecture search for this specific disease, leaving practitioners without evidence-based model selection guidance.

4. Absence of TinyML-Optimized Models for Offline Field Use:

Existing disease detection apps require cloud connectivity for inference. Sri Lanka's major onion-growing regions in the Dry Zone have unreliable internet connectivity. No plant disease detection model has been specifically optimized using TinyML techniques (INT8 quantization, pruning) to achieve sub-10 MB deployment for offline inference on basic Android smartphones commonly used by Sri Lankan farmers.

5. No Integration with Multilingual Advisory Systems:

Current disease detection tools provide diagnostic outputs in English without context-aware treatment recommendations. No existing system integrates CNN-based disease classification with a multilingual advisory pipeline capable of explaining diagnoses and recommending fungicides in Sinhala or Singlish.

Evidence-Based Justification for Novelty:

Feature	Plant.id	Plantix	Leafsnap	PlantVillage Models	Proposed System
Sri Lankan Onion-Specific Training	X	X	X	X	✓
Purple Blotch Severity Grading (4-level)	X	X	X	X	✓
AutoML Architecture Optimization	X	X	X	X	✓
TinyML Offline Deployment (<10 MB)	X	X	X	X	✓
Severity Score (0–100%)	X	X	X	X	✓
Fungicide & Cultural Control Recommendations	X	Partial	X	X	✓
Grad-CAM Explainability	X	X	X	X	✓
Multilingual Advisory Integration	X	X	X	X	✓

Table 1: Comparison of Existing Plant Disease Detection Systems

2. OBJECTIVE

2.1 Main Objective

To design, develop, and validate a high-accuracy, mobile-optimized

CNN-based system for real-time detection and severity classification of purple blotch disease (*Alternaria porri*) in onion crops under Sri Lankan field conditions, achieving >90% classification accuracy on a locally collected dataset and deployable as a sub-10 MB TensorFlow Lite model for offline inference on smartphones.

2.2 Specific Objectives

1. **Collect and Preprocess a Locally Representative Disease Dataset:** Collect 3,000+ high-resolution onion leaf images from FCRDI demonstration plots in Maha Illuppallama, spanning four severity categories (Healthy, Early, Mid, Severe) and mixed disease/pest damage, with expert pathologist labeling and verification.
2. **Implement and Compare Lightweight CNN Architectures:** Evaluate EfficientNet-B0/B1, MobileNetV3, and Vision Transformer (ViT-Small) architectures using transfer learning from ImageNet, with comprehensive data augmentation (rotation, zoom, noise, lighting variation) to maximize generalization on limited field data.
3. **Automate Architecture Search Using AutoML (FLAML):** Deploy FLAML to systematically search across candidate architectures, hyperparameters, and training configurations, creating an extensible framework that can be adapted for pest and nutrient detection modules.
4. **Optimize for Mobile Deployment Using TinyML Techniques:** Apply post-training INT8 quantization and magnitude-based pruning to reduce the best-performing model to <10 MB (TFLite format) while maintaining >90% accuracy, enabling offline inference on resource-constrained Android devices.
5. **Develop Severity Scoring and Treatment Recommendation Pipeline:** Implement a severity scoring system (0–100%) based on classification confidence and lesion extent analysis, linked to a recommendation engine providing crop-specific fungicide prescriptions and cultural control practices.

6. **Validate Through Field Testing with Farmers:** Conduct field validation with onion farmers at FCRDI training sessions, measuring classification accuracy, response latency, user satisfaction, and real-world diagnostic agreement with expert pathologist assessments.
-

3. METHODOLOGY

Research Approach

This research follows a **Design Science Research (DSR)** methodology, which emphasizes the iterative design, development, and evaluation of an IT artifact (the CNN-based disease detection model) to solve an identified real-world problem. The DSR framework guides the process through problem identification, solution design, development, demonstration, and evaluation cycles [16].

Technical Methods and Algorithms

1. Data Collection and Preprocessing:

- **Source:** FCRDI onion demonstration plots, Maha Illuppallama
- **Image Capture:** High-resolution leaf images (minimum 1024×768 pixels) using smartphone cameras under natural field conditions
- **Categories:** Healthy (500+), Early purple blotch (700+), Mid-stage disease (800+), Severe infection (700+), Mixed damage (300+)
- **Preprocessing Pipeline:**
 - Leaf segmentation using GrabCut / U-Net to isolate leaf regions from complex field backgrounds
 - Artifact removal (soil, water droplets, shadows)
 - Image resizing and normalization (224×224 for EfficientNet/MobileNet, 384×384 for ViT)
 - Data augmentation: random rotation ($\pm 30^\circ$), horizontal/vertical flip, zoom (0.8–1.2×), brightness/contrast variation, Gaussian noise injection

2. Model Architectures and Transfer Learning:

Three candidate architectures will be evaluated:

- **EfficientNet-B0/B1:** Compound-scaled CNN with depthwise separable convolutions, squeeze-and-excitation blocks. Pre-trained on ImageNet, final classification head replaced with 4-class softmax layer.
- **MobileNetV3-Large:** Lightweight architecture optimized via neural architecture search. Inverted residual blocks with h-swish activation. Pre-trained on ImageNet, fine-tuned on disease dataset.
- **Vision Transformer (ViT-Small):** Patch-based self-attention architecture. Image divided into 16×16 patches, processed through 6-layer transformer encoder. Pre-trained on ImageNet-21K.

Common training configuration:

- **Loss Function:** Cross-entropy with label smoothing ($\epsilon = 0.1$)
- **Optimizer:** AdamW with cosine learning rate scheduling
- **Initial Learning Rate:** $1e-4$ (warm-up for 5 epochs)
- **Batch Size:** 32
- **Epochs:** 50 with early stopping (patience = 10)
- **Regularization:** Dropout (0.3), weight decay ($1e-4$)

3. AutoML Architecture Search (FLAML):

FLAML will automate the search across:

- Architecture selection (EfficientNet-B0, B1, MobileNetV3, ViT-Small)
- Learning rate, batch size, dropout rate, weight decay
- Augmentation strategy combinations
- Evaluation using 5-fold stratified cross-validation with accuracy and F1 as selection criteria

4. TinyML Optimization:

The best-performing model undergoes:

- **Post-Training INT8 Quantization:** Weights and activations quantized from FP32 to INT8, reducing model size by $\sim 4\times$
- **Magnitude-Based Pruning:** Iterative pruning of weights below

threshold, followed by fine-tuning to recover accuracy

- **Target:** Model size <10 MB, inference latency <500 ms on mid-range Android device

5. Severity Scoring and Recommendation Engine:

- Severity score (0–100%) derived from softmax confidence distribution across severity classes
- Lesion extent estimation using Grad-CAM activation area analysis
- Rule-based recommendation engine mapping severity levels to fungicide prescriptions and cultural practices sourced from FCRDI guidelines

Tools and Platforms

Tool/Platform	Purpose	Justification
TensorFlow / Keras	Model training and evaluation	Industry-standard, excellent TFLite conversion support
FLAML	AutoML architecture search	Lightweight, fast, minimal computational overhead
TensorFlow Lite	Mobile model deployment	Native Android support, INT8 quantization tools
FastAI	Rapid prototyping and augmentation	High-level API for efficient experimentation
Flutter	Mobile application development	Cross-platform (Android/iOS), single codebase
Flask/FastAPI	API backend development	Lightweight, Python-native, ideal for ML model serving
Firebase	Real-time database & authentication	Scalable, free tier available, mobile-optimized
LabelStudio	Image annotation	Open-source, supports multi-class image labeling
Git/GitHub	Version control	Industry standard for collaborative development

Grad-CAM (tf-explain)	Model explainability	Visual explanations for disease region identification
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Data Collection and Ethical Considerations

Data Sources:

- **FCRDI Demonstration Plots:** Primary source of onion leaf images collected under natural field conditions across multiple growth stages
- **Expert Labeling:** FCRDI plant pathologists verify diagnosis and assign severity levels for each image
- **Supplementary Data:** Environmental metadata (temperature, humidity, growth stage) recorded during image collection

Data Format: 3,000+ high-resolution leaf images across 5 categories:

- Healthy onion leaves (500+)
- Early purple blotch symptoms (700+)
- Mid-stage disease progression (800+)
- Severe infection (700+)
- Mixed disease/pest damage (300+)

Image Labeling Protocol:

- Primary labeling by trained student researchers following FCRDI severity classification guide
- Verification and correction by Dr. W.M.K Fernando (domain expert)
- Inter-annotator agreement (Cohen's Kappa) computed on 10% sample subset

Ethical Considerations:

- This project requires ethical clearance (submitted to Research Ethics Committee)
- Farm owner consent obtained for all on-farm photography
- All farmer data anonymized; no personally identifiable information stored

- Image metadata stripped of GPS coordinates before storage
- Data management plan with confidentiality protocols
- Compliance with Sri Lankan data protection regulations

Validation Strategy and Performance Metrics

Metric	Target	Method
Classification Accuracy	≥ 90%	5-fold stratified cross-validation on test set
F1 Score (macro-averaged)	≥ 88%	Precision-recall analysis per severity class
Inference Latency	< 500 ms	Benchmarking on mid-range Android device (Snapdragon 665)
Model Size (TFLite)	< 10 MB	Post-quantization and pruning measurement
Accuracy Drop Post-Quantization	< 2%	Comparison of FP32 vs. INT8 model accuracy
Grad-CAM Agreement with Expert	≥ 80%	Expert review of highlighted disease regions
Field Diagnostic Agreement	≥ 85%	Comparison with pathologist diagnosis on 200 field samples

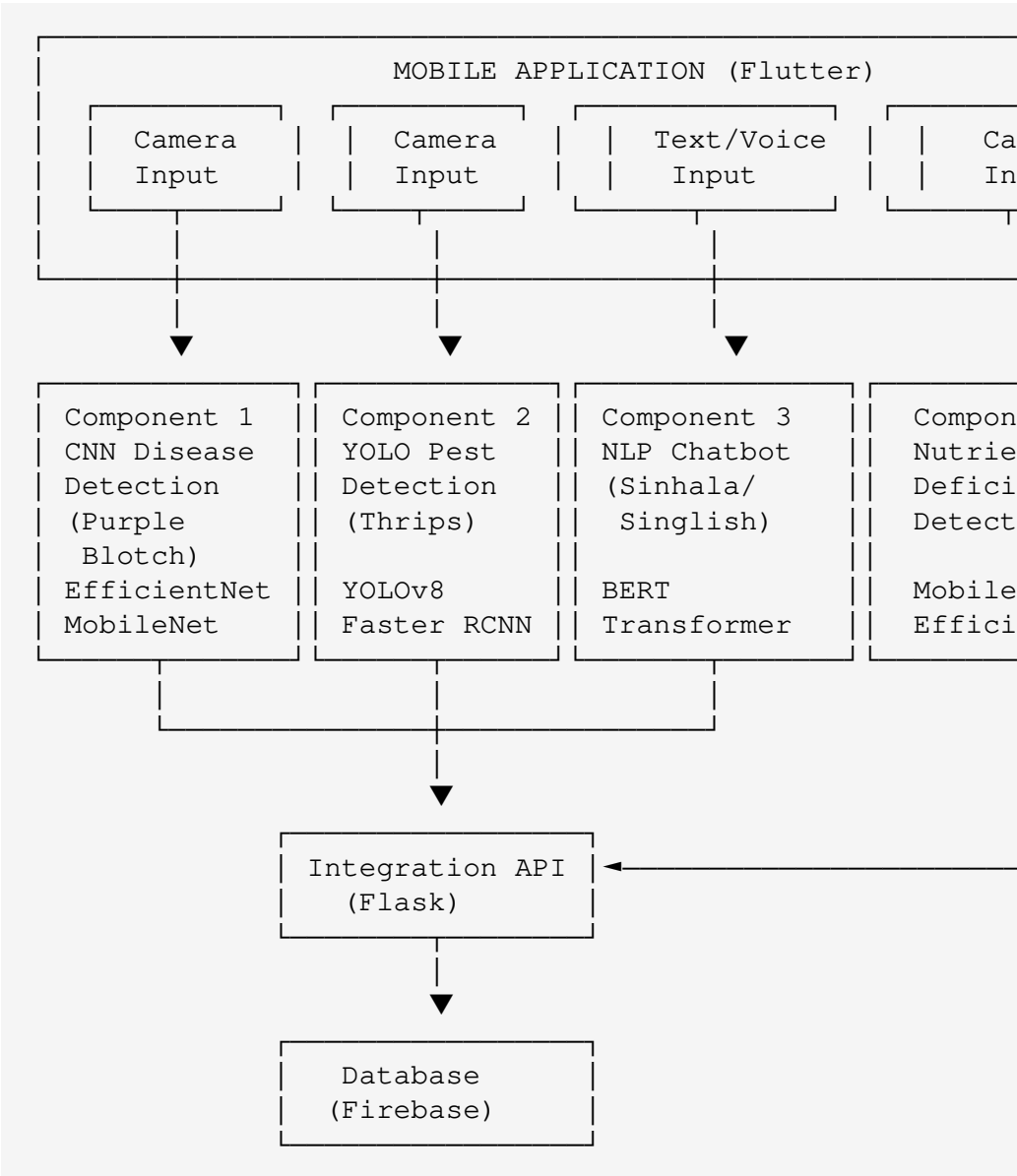
4. HIGH-LEVEL SYSTEM ARCHITECTURE

Overview of the Proposed Solution

The Automated Diagnostic Framework for Onion Crop Health Monitoring is a mobile-first AI platform comprising four integrated components. This section describes the overall system and highlights the CNN-Based

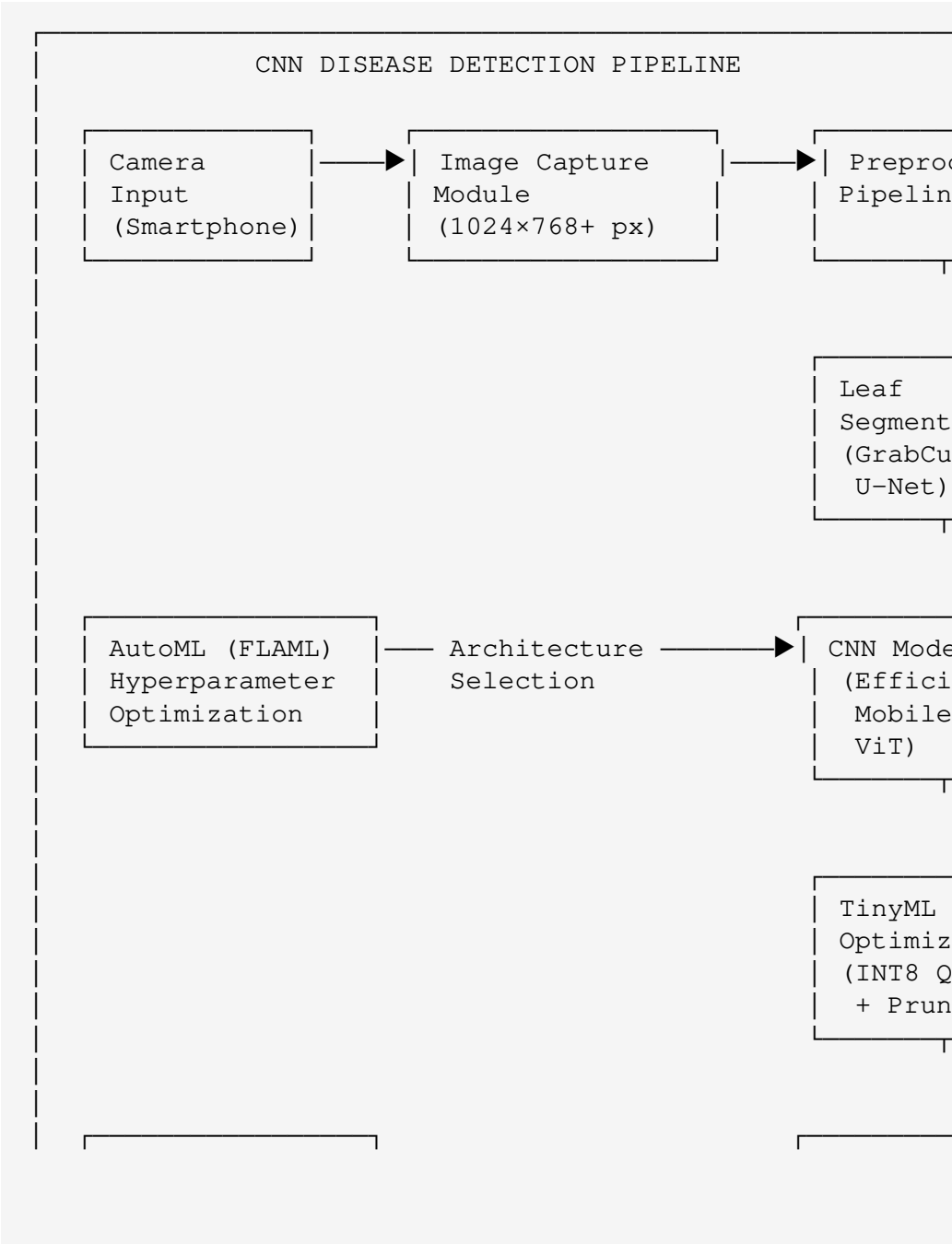
Purple Blotch Disease Detection (Component 1) within the larger architecture.

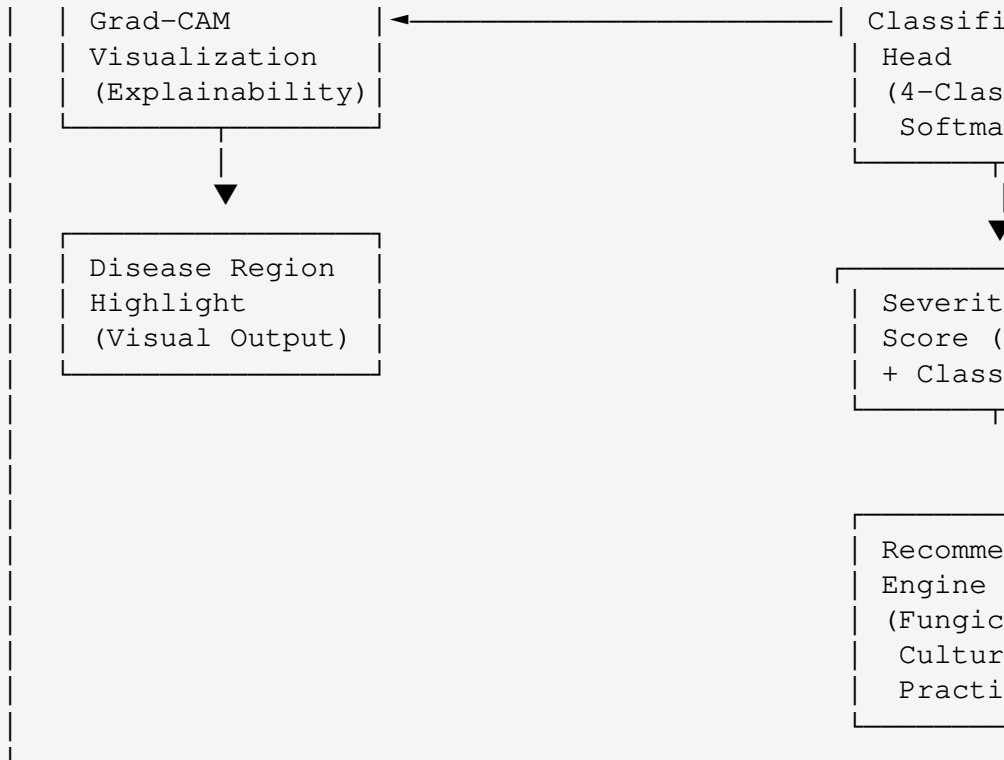
System Diagram of Components



Detailed Disease Detection Component Architecture

(Component 1)





Component Interactions

The disease detection module (Component 1) interacts with other system components through well-defined REST APIs:

- **Component 1 → Component 3:** When the CNN model identifies purple blotch, it sends the severity classification (Healthy/Early/Mid/Severe), confidence score, severity percentage (0–100%), and Grad-CAM visualization to the chatbot API. The chatbot translates this into a Sinhala/Singlish advisory message explaining the diagnosis and recommended fungicide application.
- **Component 1 → Component 4:** Disease severity information is shared with the nutrient deficiency module to differentiate overlapping symptoms (e.g., leaf discoloration caused by disease vs. nutrient stress).

- **Component 1 → Integration API:** Classification results, severity scores, timestamps, and environmental metadata are stored via the Flask API for historical tracking and model improvement.

Data Flow

1. Farmer captures an image of the onion leaf using the mobile app camera
2. Image is preprocessed: resized, normalized, and leaf region segmented
3. The TFLite model performs inference locally on the device (offline capable)
4. Classification result (Healthy/Early/Mid/Severe) and severity score are generated
5. Grad-CAM visualization highlights the disease-affected regions
6. Recommendation engine maps severity to fungicide prescription and cultural control practices
7. Results are displayed to the farmer and optionally sent to the chatbot for multilingual explanation

Scalability and Security Considerations

- **Scalability:** The TFLite model runs entirely on-device, eliminating server-side scaling concerns for inference. The AutoML framework creates an extensible architecture search pipeline reusable for pest and nutrient detection modules. The training pipeline supports incremental learning as new data is collected.
 - **Security:** All API communications use HTTPS encryption. Farmer image data is anonymized at the point of collection with GPS metadata stripped. Firebase authentication ensures secure access control. The system complies with Sri Lankan data protection requirements.
-

5. USER REQUIREMENTS

Stakeholder Identification

Stakeholder	Role	Primary Need
Onion Farmers	End Users	Rapid, accurate disease identification with treatment advice
FCRDI Field Officers	Advisory Support	Validate diagnostic accuracy, monitor disease trends
Plant Pathologists	Domain Experts	Ensure diagnostic accuracy matches expert assessment
System Administrators	Technical Support	Manage model updates and performance monitoring

Functional Requirements

ID	Requirement	Priority
FR-01	The system shall accept onion leaf images captured via smartphone camera	High
FR-02	The system shall perform leaf segmentation to isolate the leaf region from background	High
FR-03	The system shall classify leaf images into four categories: Healthy, Early, Mid, Severe	High
FR-04	The system shall generate a severity score (0–100%) for each classification	High
FR-05	The system shall provide Grad-CAM visualizations highlighting disease-affected regions	High
FR-06	The system shall recommend appropriate fungicides and cultural control practices based on severity	High
FR-07	The system shall perform inference offline using the on-device TFLite model	High

FR-08	The system shall send classification results to the chatbot API for multilingual explanation	Medium
FR-09	The system shall log diagnostic results for historical tracking (anonymized)	Medium
FR-10	The system shall support model updates via over-the-air (OTA) distribution	Medium
FR-11	The system shall display inference confidence alongside classification results	Medium
FR-12	The system shall support batch image processing for multiple leaves	Low

Table 2: Functional Requirements

Non-Functional Requirements

ID	Requirement	Category	Target
NFR-01	Classification accuracy shall exceed 90% on test set	Accuracy	$\geq 90\%$
NFR-02	Inference latency shall be under 500 ms on target device	Performance	$< 500 \text{ ms}$
NFR-03	TFLite model size shall not exceed 10 MB	Efficiency	$\leq 10 \text{ MB}$
NFR-04	Accuracy loss after INT8 quantization shall be under 2%	Accuracy	$< 2\%$
NFR-05	The UI shall be intuitive for users with limited digital literacy	Usability	SUS ≥ 70
NFR-06	All API communications shall use HTTPS encryption	Security	TLS 1.2+

NFR-07	Farmer image data shall be anonymized and stripped of GPS metadata	Privacy	Compliance
NFR-08	The system shall function fully offline for core disease detection	Availability	Offline mode

Table 3: Non-Functional Requirements

6. COMMERCIALIZATION PLAN

Target Market and Customer Persona

Primary Market: Sri Lanka's 150,000+ onion farming households concentrated in the Dry Zone (Maha Illuppallama, Matale, Dambulla, Jaffna).

Customer Persona:

- **Name:** Saman (representative farmer)
- **Age:** 35–55 years
- **Education:** O/L or below
- **Language:** Sinhala, Singlish
- **Technology:** Owns a basic Android smartphone (Snapdragon 665 or equivalent), limited data plan
- **Pain Point:** Cannot identify purple blotch early enough; relies on neighbors' advice or waits for government officer visits; often applies wrong fungicide or applies it too late

Value Proposition

A single mobile application that allows farmers to photograph their onion leaves, receive instant AI-powered disease diagnosis with severity grading, and get specific fungicide recommendations—all working offline

without internet connectivity, replacing the need for expert visits that may take days or weeks.

Revenue Model

The system follows a **Freemium** approach:

Tier	Features	Price
Free	Basic disease detection, classification result, general advice	Free
Premium	Severity scoring, Grad-CAM visualization, detailed fungicide recommendations, historical tracking, chatbot integration	LKR 300/ month per farmer
Enterprise	Bulk licensing for agricultural cooperatives, white-label options, analytics dashboard, regional disease outbreak monitoring	Custom pricing

Cost Estimation

Item	Estimated Annual Cost
Cloud infrastructure (GCP – training only)	LKR 96,000
App Store + Play Store hosting	LKR 38,454
Data storage (Firebase + GCS)	LKR 12,000
GPU compute for model retraining	LKR 48,000
Marketing and farmer outreach	LKR 50,000
Total	LKR 244,454

Competitive Advantage

1. **First *Alternaria porri*-specific AI model** trained on Sri Lankan

- onion leaf images
2. **Four-level severity classification** with quantitative scoring (0–100%)
 3. **Fully offline operation** via TinyML-optimized TFLite model (<10 MB)
 4. **Grad-CAM explainability** showing which leaf regions drove the diagnosis
 5. **AutoML-optimized architecture** ensuring best accuracy-efficiency tradeoff
 6. **FCRDI-validated** disease labeling and fungicide recommendations
 7. **Integration with multilingual chatbot** for Sinhala/Singlish advisory

Intellectual Property Considerations

- The fine-tuned CNN model and locally collected onion disease dataset constitute protectable intellectual property
- The AutoML pipeline configuration and TinyML optimization methodology may be published as research contributions
- Open-source components (TensorFlow, FLAML, Flutter) are used under their respective licenses (Apache 2.0, BSD)

7. BUDGET AND JUSTIFICATION

Item	Description	Cost (LKR)	Occurrence	Justification
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Cloud GPU Instance	GCP Compute Engine (T4 GPU) for model training	8,000	Monthly	GPU-accelerated training required for EfficientNet/ViT fine-tuning with data augmentation
Firebase (Realtime DB + Auth)	Database hosting and user authentication	0 (Free tier)	Monthly	Free tier sufficient for initial deployment (1GB storage, 10K auth/month)
Play Store Developer Account	Publishing the mobile app on Google Play	7,754	One-Time	Required for Android app distribution
App Store Developer Account	Publishing on Apple App Store	30,700	Annual	Required for iOS app distribution
Cloud Storage (GCS)	Storing training images, model weights, logs	0.023 USD/GB	Monthly	Cost-effective cloud storage for 3,000+ images and model artifacts
Domain Name	Custom domain for API endpoints	3,000	Annual	Professional branding and consistent API access

Field Visit Transportation	Travel to FCRDI sites for image collection	15,000	One-Time	Multiple visits required for seasonal image collection across growth stages
Development Hardware	No additional cost (university lab resources)	0	—	

Table 4: Budget Breakdown and Justification

Total Estimated Cost (Year 1): ~LKR 180,000

8. WORK BREAKDOWN STRUCTURE (WBS)

Task Breakdown – Component 1 (CNN Disease Detection – IT22054890)

WBS ID	Task	Duration	Deliverable
1.1	Literature review and research methodology design	Weeks 1–3	Literature review report
1.2	Field visits to FCRDI; image collection protocol design	Weeks 3–6	Image collection protocol document
1.3	Collect 3,000+ onion leaf images from FCRDI demonstration plots	Weeks 4–12	Annotated image dataset

1.4	Expert labeling and verification by FCRDI pathologists	Weeks 8–14	Validated labeled dataset
1.5	Image preprocessing pipeline: segmentation, augmentation, normalization	Weeks 10–16	Preprocessing pipeline code
1.6	Implement EfficientNet, MobileNet, ViT architectures with transfer learning	Weeks 14–22	Trained baseline models
1.7	AutoML architecture search using FLAML	Weeks 20–26	Optimal model configuration
1.8	TinyML optimization: INT8 quantization and pruning	Weeks 24–30	TFLite model (<10 MB)
1.9	Severity scoring and recommendation engine development	Weeks 28–34	Severity pipeline + recommendation module
1.10	Grad-CAM explainability integration	Weeks 30–34	Visual explanation module
1.11	Mobile integration and on-device testing	Weeks 32–38	Mobile-deployed detection system
1.12	Cross-component API integration (with Components 2, 3, 4)	Weeks 34–40	Integrated API endpoints
1.13	Field validation with farmers and pathologists	Weeks 38–44	Field validation report
1.14	Iterative refinement based on field feedback	Weeks 42–46	Optimized final system
1.15	Documentation and thesis writing	Weeks 44–48	Final dissertation

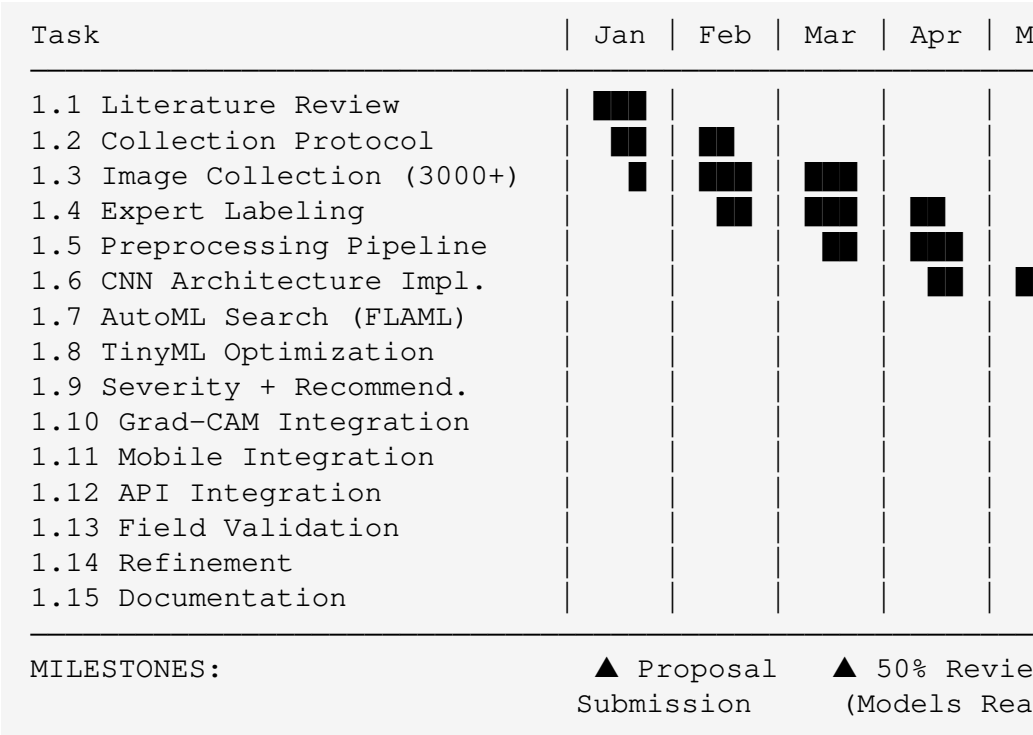
Table 5: Work Breakdown Structure

Timeline Alignment with Group

All four members follow a synchronized timeline with shared milestones at:

- **Week 12:** Data collection complete (all components)
- **Week 24:** Core models trained and individually validated
- **Week 34:** Cross-component integration complete
- **Week 42:** User testing complete
- **Week 48:** Final submission

9. GANTT CHART



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11. APPENDICES

Appendix A: Disease Severity Classification Guide

Severity Level	Visual Symptoms	Severity Score Range	Recommended Action
Healthy	No visible lesions; uniform green coloration; vigorous growth	0–10%	Continue regular monitoring; maintain preventive spray schedule

Early	Small, water-soaked spots; initial purple-brown discoloration on leaf tips; lesions < 5mm	11–35%	Apply Mancozeb 80% WP at 2g/L immediately; repeat at 7-day intervals
Mid	Expanding concentric ring lesions (5–20mm); purple-brown coloration with yellow halo; multiple leaves affected	36–65%	Apply Chlorothalonil 75% WP at 2g/L or Tebuconazole 25% EC at 1ml/L; remove severely affected leaves
Severe	Large coalescing lesions (>20mm); extensive leaf necrosis; leaf tip dieback; reduced plant vigor	66–100%	Apply systemic fungicide (Propiconazole 25% EC 1ml/L); consider crop salvage; plan next-season prevention

Appendix B: Image Collection Protocol

Equipment:

- Smartphone camera (minimum 12 MP, autofocus enabled)
- Color reference card (X-Rite ColorChecker)
- Centimeter ruler for scale reference

Collection Procedure:

1. Select representative plants from designated FCRDI plots
2. Photograph individual leaves at 30 cm distance, perpendicular to leaf surface
3. Include healthy and diseased portions in same frame where possible
4. Capture images under natural sunlight (avoid harsh shadows; 8 AM–4 PM)

- 5. Record metadata: date, time, plot ID, growth stage, recent weather conditions
- 6. Capture minimum 5 images per plant from different angles
- 7. Place color reference card adjacent to leaf for white balance calibration

Quality Control:

- Discard blurred, overexposed, or underexposed images
- Minimum resolution: 1024 × 768 pixels
- Review 10% of images with pathologist within 48 hours of collection

Appendix C: Model Architecture Comparison Summary

Architecture	Parameters	Model Size (FP32)	ImageNet Top-1 Accuracy	Mobile Inference	Strengths
EfficientNet-B0	5.3M	21 MB	77.1%	✓ Fast	Best accuracy-efficiency tradeoff; compound scaling
EfficientNet-B1	7.8M	31 MB	79.1%	✓ Moderate	Higher accuracy at moderate cost
MobileNetV3-Large	5.4M	22 MB	75.2%	✓ Fastest	NAS-optimized for mobile; h-swish activation

ViT-Small	22M	88 MB	81.4%	✗ Slow	Global context; strong with augmentation

Appendix D: Risk Analysis Matrix

Risk	Probability	Impact	Mitigation
Insufficient disease image diversity	Medium	High	Collect across multiple growth stages, seasons, and plots; extensive data augmentation
Class imbalance (fewer early-stage samples)	High	Medium	Oversampling (SMOTE for features), weighted loss function, augmentation focus on minority classes
ViT overfitting on small dataset	Medium	Medium	Use smaller ViT variant, aggressive augmentation, early stopping, compare with CNN baselines
Accuracy degradation after INT8 quantization	Low	High	Quantization-aware training (QAT) if post-training quantization exceeds 2% accuracy loss
Field images differ from training data distribution	High	High	Include field-captured images in training; domain adaptation techniques; continuous data collection

Seasonal variation in disease appearance	Medium	Medium	Collect images across Maha and Yala seasons; include environmental metadata as auxiliary features
Confusion with similar foliar diseases (Stemphylium blight)	Medium	High	Add "Other Disease" class; consult pathologist for borderline cases; Grad-CAM analysis for verification

Appendix E: Ethical Clearance Documents

(To be attached)

- Informed consent form (Sinhala version) for farm owners
- Image collection consent form
- Data management plan
- Confidentiality agreement template
- Ethics committee application form (submitted January 2026)